-Text data requires preprocessing to be analyzed and extract meaningful information and removing unnecessary words and redundant expression.

-We need to transform unstructured data into structured data, by converting them into a vector or a matrix form to apply machine learning algorithm based on vector space.

- Bag of words represents documents as a collection of words, where a text is represented in a vector of an unordered collection of words. We consider words as atomic symbols, represented in the discrete space.

-Bag of words lets us represent a document as a Term-document matrix. It's hard to extract context information since only term frequency is analyzed. It's not possible to reconstruct the original text based on the term-document matrix.

-In bag of words each word becomes a feature, increasing the size of the matrix. This is why it's important to unnecessary information, by removing the difference between lower-cased and upper-cased words, removing punctuation and numbers.

-Removing stop words (words that don't carry information) helps the machine learning algorithm to perform better. SMART and "MySQL stop word list" are stop word lists having more than 500 common English stop words in it.

-Term frequency is the number of times that a term t occurs in a document (TF), while document frequency is the number of documents in which a term t appears (DF). We should give high weight for rare terms than common terms.

-Inverse document frequency (IDF) evaluates how rare a specific term is across the entire document corpus. Idf = log10(N/dft). N = total number of doc dft = number of doc with the term t.

-TF-IDF(W) = tf(w) x log(N/df(w)) this formula is used for text mining and information retriever.

-Most frequently used TF-IDF formulas used:

Term frequency		Document frequency		Normalization	
n (natural)	$tf_{t,d}$	n (no)	1	n (none)	1
l (logarithm)	$1 + \log(tf_{t,d})$	t (idf)	$\log \frac{N}{df_t}$	c (cosine)	$\frac{1}{\sqrt{w_1^2+w_2^2+\ldots+w_M^2}}$
a (augmented)	$0.5 + \frac{0.5 \times tf_{t,d}}{\max_t(tf_{t,d})}$	p (prob idf)	$\max\{0, \log \frac{N-df_t}{df_t}\}$	u (pivoted unique)	$1/u$
b (boolean)	$\begin{cases} 1 & \text{if } tf_{t,d} > 0 \\ 0 & \text{otherwise} \end{cases}$			b (byte size)	$1/CharLength^{\alpha}$, $\alpha < 1$
L (log ave)	$\frac{1+\log(tf_{t,d})}{1+\log(ave_{t \in d}(tf_{t,d}))}$				

-N-Gram based language models in NLP uses the N-1 word in a sequence to predict the next word.

-Bigram analysis helps understanding the structure and the context of a document by analyzing the frequency of 2 words pairing and treating them in some cases like a single word.